PCT/PCM4: The Noise Line-Up

Item 4: Noise Histogram Interpretation
The salt-and-pepper panel is identified by sharp spikes near the extreme dark and bright intensity values (near 0 and 255), because the noise creates random black and white pixels. The Gaussian panel is identified by a broad, bell-shaped histogram centered around the underlying intensity, because Gaussian noise produces continuous variations around the mean.

Item 5: Gaussian Noise Calculation
Given zero-mean Gaussian noise with $\sigma = 20$ and an 8-bit peak value of 255:

$MSE = \sigma^2 = 20^2 = 400$

$PSNR = 10 \log10(255^2 / 400) \approx 22.1$ dB

Therefore, MSE = 400 and PSNR = 22.1 dB.

Item 6: Prediction: Poisson Noise
Poisson noise will generally have greater absolute variation in bright regions because its noise variance increases with the photon count. This differs from the Gaussian noise in Item 5, where $\sigma = 20$ is constant, so the noise variance is approximately the same for dark and bright regions.